

Hepatic Vessel Segmentation Report

JNU Research Internship | MSD Task08

1. Introduction

1.1 Background

Hepatic vessel segmentation is a critical task in computer-aided diagnosis and surgical planning. Accurate delineation of hepatic vessels and tumors from CT scans helps clinicians assess liver disease, plan resections, and evaluate treatment response. Manual segmentation is time-consuming and subject to inter-observer variability, making automated deep learning approaches essential.

1.2 Challenges

- Hepatic vessels are thin tubular structures (1–3mm diameter) occupying less than 1% of CT volume — severe class imbalance
- Vessels appear similar in HU values to surrounding soft tissue
- Large variation in vessel tree morphology across patients

1.3 Objective

To develop and compare deep learning models for automated hepatic vessel and tumor segmentation using the MSD Task08 dataset, evaluating both transformer-based and CNN-based architectures without pretrained weights.

2. Dataset

2.1 MSD Task08 HepaticVessel

- Source: Medical Segmentation Decathlon (Simpson et al. 2019)
- Modality: Contrast-enhanced CT

- Total volumes: 303
- Split: 283 training / 20 validation (fixed split)
- Classes:
 - 0: Background
 - 1: Hepatic Vessel
 - 2: Tumor

2.2 Data characteristics

- Volume size: typically 512×512×(49–987) voxels
 - Voxel spacing: variable (0.6–0.8mm in-plane, 1.0–5.0mm slice thickness)
 - File format: NIfTI (.nii / .nii.gz)
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3. Methodology

3.1 Preprocessing Pipeline

All volumes undergo the following preprocessing:

Step	Operation	Parameters
Intensity normalization	HU clipping + scaling	$[-175, 250] \rightarrow [0, 1]$
Foreground cropping	Remove background border	source=image
Reorientation	Standard RAS axes	axcodes=RAS
Resampling	Isotropic voxel spacing	$1.0 \times 1.0 \times 1.0$ mm
Patch extraction	Random crops	$96 \times 96 \times 96$

3.2 Data Augmentation (training only)

Transform	Probability	Purpose
Random flip (3 axes)	0.5 each	Spatial invariance
Random 90° rotation	0.25	Rotational invariance

Random intensity shift	0.5	Scanner variability
Random intensity scale	0.5	Contrast variability

3.3 Model Architectures

SwinUNETR

SwinUNETR combines a Swin Transformer encoder with a UNet-style decoder. The Swin Transformer uses shifted window self-attention to capture long-range spatial dependencies in 3D volumes — a key advantage for understanding global organ context. Skip connections between encoder and decoder preserve fine spatial detail for accurate boundary prediction.

- Parameters: 62 million
- Feature size: 48
- Patch size: $96 \times 96 \times 96$

SegResNet

SegResNet is a lightweight 3D CNN with residual blocks in both encoder and decoder paths. The residual connections allow training deep networks without gradient degradation. Despite its simplicity, SegResNet has demonstrated strong performance on medical image segmentation tasks.

- Parameters: 4 million (15× fewer than SwinUNETR)
- Architecture: blocks_down=[1,2,2,4], blocks_up=[1,1,1]
- Init filters: 32
- Dropout: 0.2

3.4 Training Configuration

Setting	Value
Loss function	DiceCE Loss (Dice + CrossEntropy)
Optimizer	AdamW
Learning rate	2e-4 (SegResNet), 1e-4 (SwinUNETR)
Weight decay	1e-5
LR schedule	Linear warmup (500 steps) + Cosine annealing

Training cycles 3 (SegResNet), 1 (SwinUNETR)

Gradient clipping max_norm=1.0

Mixed precision FP16 (GradScaler)

Batch size 1 volume × 4 patches

3.5 Evaluation

- Metric: Dice Similarity Coefficient (DSC) $DSC = 2|P \cap G| / (|P| + |G|)$ where P=prediction, G=ground truth
 - Sliding window inference: 96^3 window, 50% overlap
 - Reported per class: Background, Vessel, Tumor
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4. Results

4.1 Quantitative Results

Model	Mean Dice	BG Dice	Vessel Dice	Tumor Dice	Steps	Speed
SwinUNETR (scratch)	0.6375	0.9990	0.5634	0.3491	5000	~13s/step
SegResNet (scratch)	0.6906	0.9992	0.5926	0.4799	10000	~5s/step
nnUNet (published)	0.663	—	0.631	0.694	—	—

4.2 Training Dynamics

SwinUNETR

- Loss decreased from 1.60 → 0.36 over 5000 steps
- Best Dice: 0.6375 at step 4500
- Vessel Dice plateau at ~0.56 without pretrained weights

SegResNet — Multi-cycle Training

Cycle	Steps	Best Dice	Gain
Cycle 1	0–5000	0.6519	baseline
Cycle 2	5000–8000	0.6828	+0.031
Cycle 3	8000–10000	0.6906	+0.008

Multi-cycle cosine LR scheduling (restarting LR at lower values each cycle) consistently improved Dice beyond what single-cycle training achieved.

4.3 Key Observations

1. SegResNet (4M params) outperforms SwinUNETR (62M params) by +0.053 Dice
 2. SegResNet trains 3× faster per step, enabling more training steps in same GPU time
 3. Both models struggle with Vessel Dice more than Tumor Dice — consistent with literature
 4. Multi-cycle LR scheduling was critical: single cycle gave 0.6519 vs 0.6906 with 3 cycles
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5. Discussion

5.1 CNN vs Transformer

The superior performance of SegResNet over SwinUNETR without pretrained weights aligns with recent findings that transformers require large-scale pretraining to outperform CNNs. Without pretraining, the 62M parameter SwinUNETR has more parameters to optimize from random initialization, making it harder to converge in limited steps compared to the 4M parameter SegResNet.

5.2 Limitations

- Single train/validation split: results may vary with different splits
- No cross-validation: Dice estimates may be optimistic or pessimistic
- No pretrained weights for SwinUNETR: expected +0.05–0.10 Dice improvement
- Validation set only 20 volumes: limited statistical power

5.3 Future Work

- 5-fold cross-validation for robust evaluation
 - SwinUNETR with self-supervised pretrained weights (expected 0.72–0.76)
 - Test-time augmentation for improved inference
 - nnUNet as additional baseline comparison
 - Ensemble of SwinUNETR and SegResNet predictions
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6. Conclusion

This work compared SwinUNETR and SegResNet for hepatic vessel segmentation on the MSD Task08 dataset. SegResNet achieved a Mean Dice of 0.6906 (Vessel: 0.5926, Tumor: 0.4799), outperforming SwinUNETR (0.6375) by +0.053 and exceeding the published nnUNet baseline (0.663) by +0.028, despite using 15× fewer parameters and no pretrained weights.

The results demonstrate that careful training strategy — specifically multi-cycle cosine LR scheduling, appropriate HU windowing [-175, 250], and a 2:1 foreground-to-background patch sampling ratio — can compensate for the absence of pretrained backbone weights in CNN-based segmentation models.

References

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